Unit 11, Lesson 3: Constructing Scatter Plots

Benchmarks

MA.8.DP.1.1 Given a set of real-world bivariate numerical data, construct a scatter plot or a line graph as appropriate for the context.

MA.8.DP.1.2 Given a scatter plot within a real-world context, describe patterns of association.

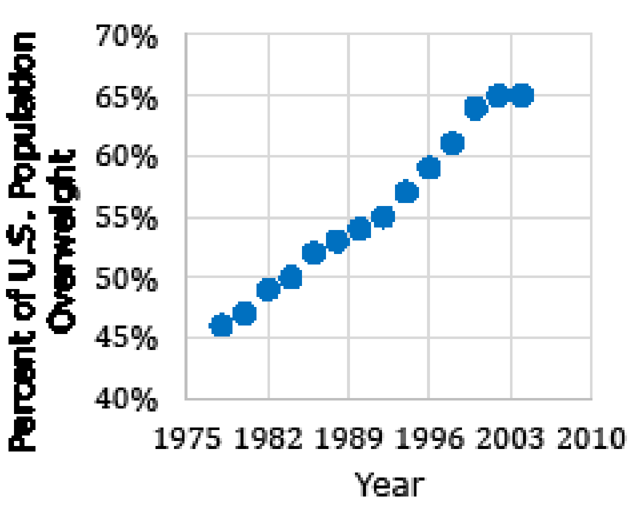
Learning Targets

- ☐ **I can describe patterns of association for the data using the words "increasing", "decreasing", "linear", or "non-linear."**
- ☐ **I can construct a scatter plot given a set of real-world data.**

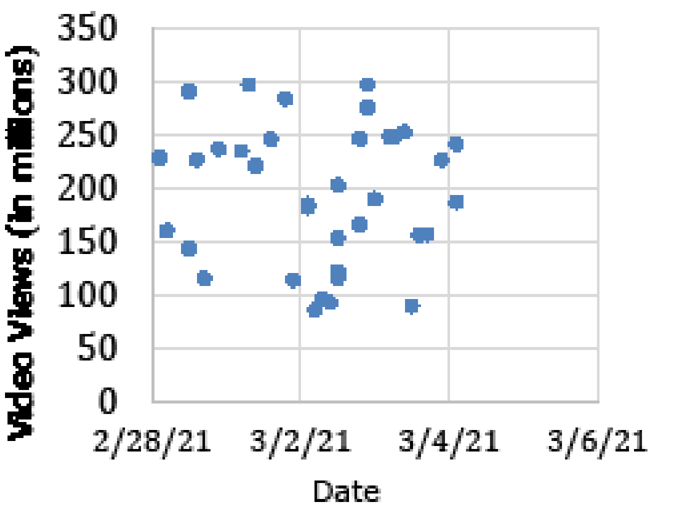
11.3.1 Warm-Up: Which One Doesn't Belong?

1. Four scatter plots are shown.

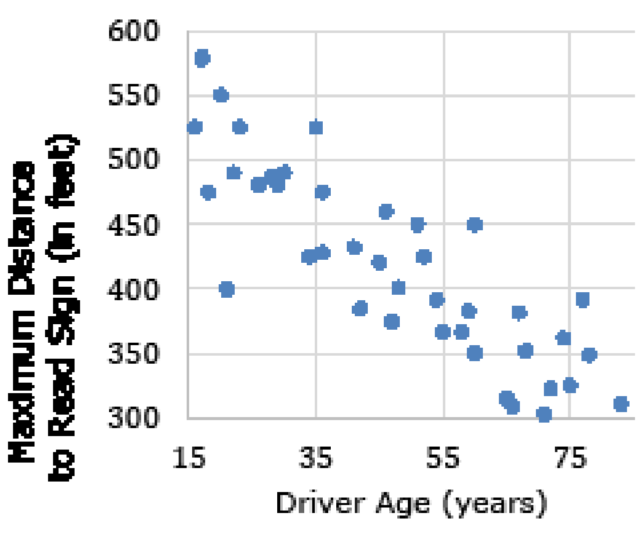
Scatter Plot A



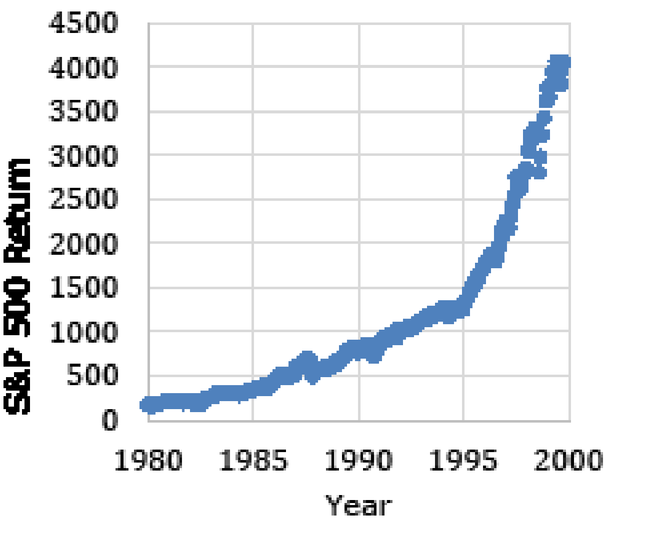
Scatter Plot B



Scatter Plot C



Scatter Plot D



a. Which one doesn't belong?

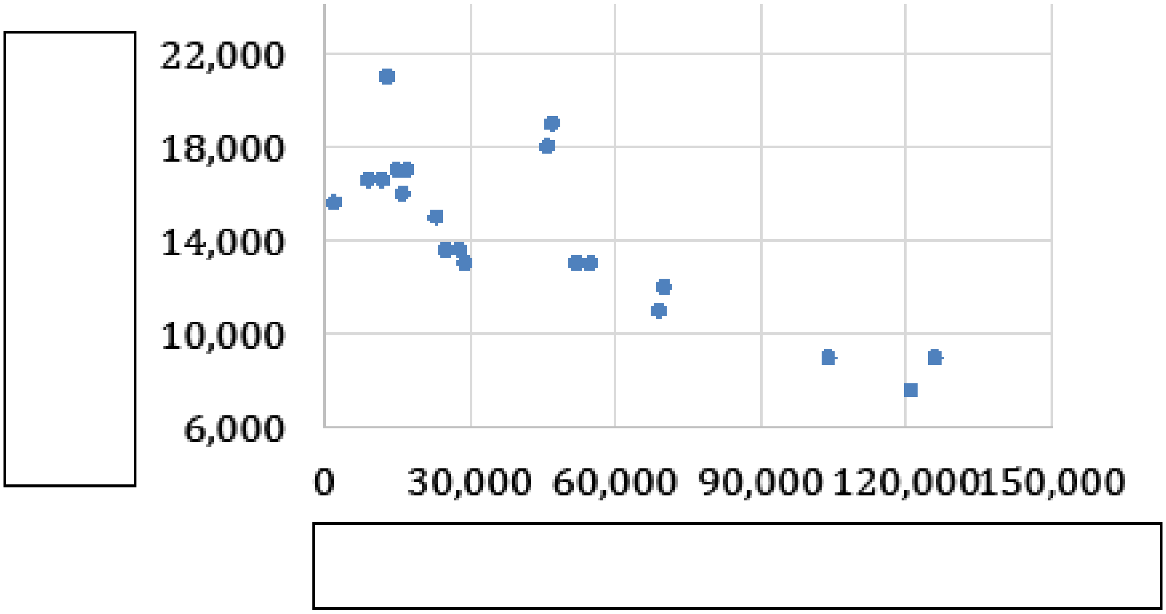
b. Justify your choice.

11.3.2 Exploration: Scatter Plots and Their Data

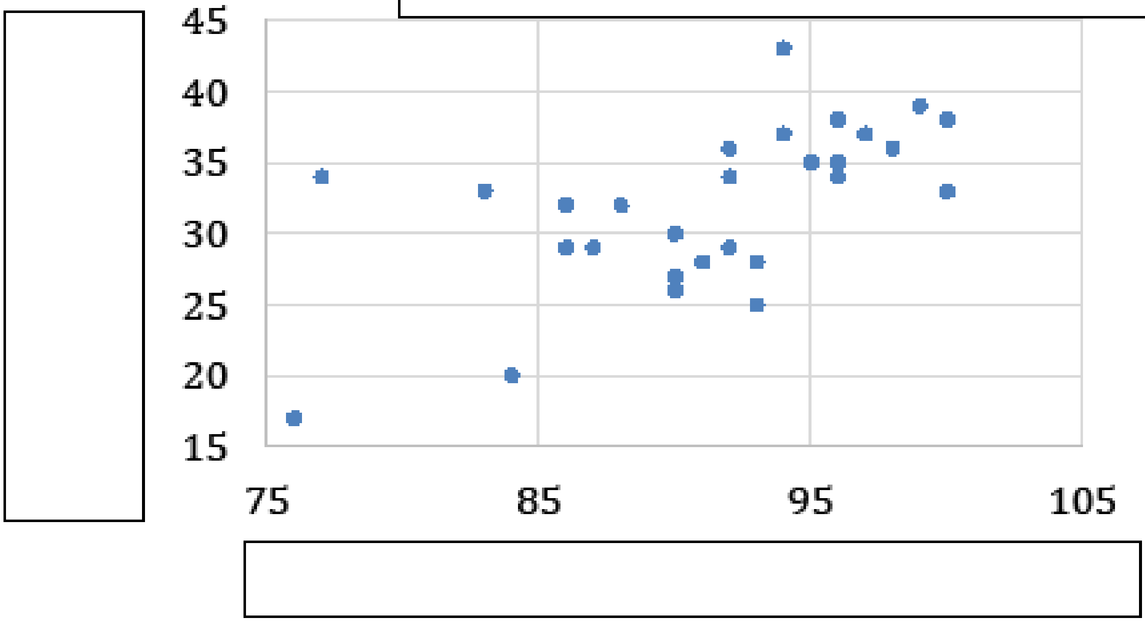
Your teacher will give you four tables of data.

1. Work with your partner to match each table with one of the scatter plots shown. Use the information from the tables to label the axes and title of each scatter plot.

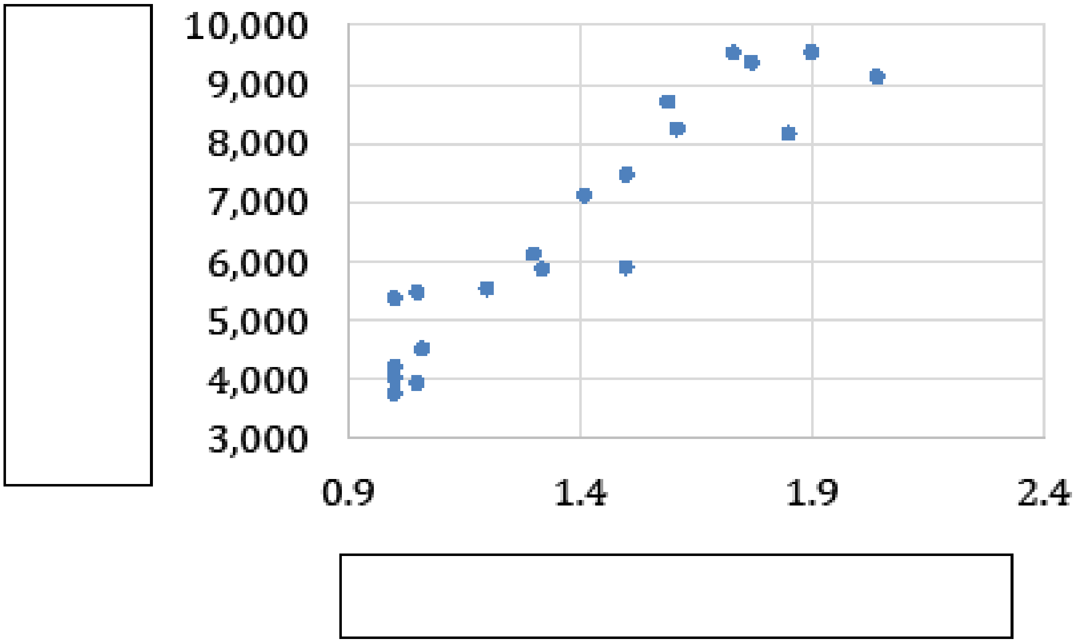
Scatter Plot I:



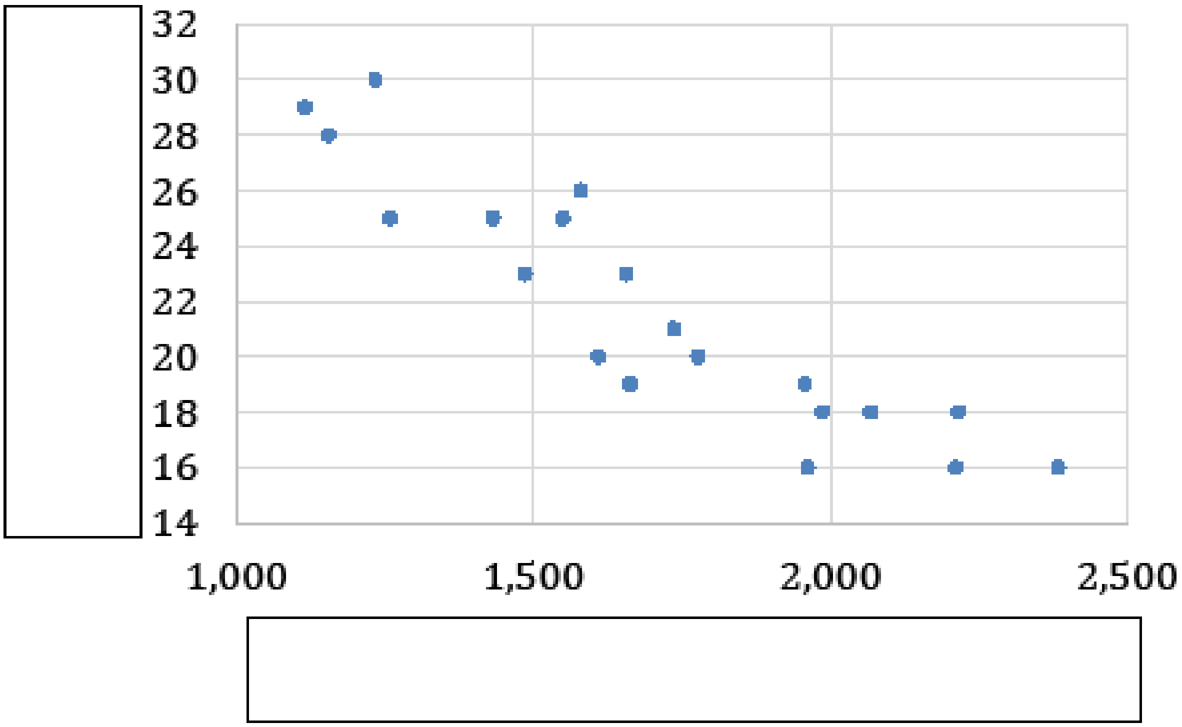
Scatter Plot II:



Scatter Plot III:



Scatter Plot IV:



2. Work with your partner to complete each statement.

a. The trend on scatter plot ____ shows that as the weight of a diamond increases, its price
- o decreases.
- o increases.

b. The trend on scatter plot ____ shows that as the mass of a new car increases, its fuel efficiency
- o decreases.
- o increases.

c. The point (76, 17) on scatter plot II represents that when the
- o daily high temperature
- o energy consumed for a home

was 76
- o degrees Fahrenheit (°F),
- o kilowatt hours (kWh),

the
- o daily high temperature
- o energy consumed for a home

was 17
- o °F.
- o kWh.

11.3.3 Bring It Together: Scatter Plots and Their Data

Each point on a scatter plot represents a specific pair of values from the raw data. The x- and y-values are used to plot each data point in the coordinate plane. When all of the points in a data set are represented on a scatter plot, trends, or patterns of ***association***, can often be seen.

An ***association*** is a way to describe the form, direction, or strength of the relationship between the two variables in a bivariate data set. For numerical data, descriptions include linear or nonlinear; positive or negative; strong or weak.

1. Refer back to the scatter plots from the Warm-Up.

   a. Using your knowledge of linear and non-linear relationships, work with your partner to complete the table.

Scatter Plot(s) with a Linear Association	☐ Scatter Plot A ☐ Scatter Plot B ☐ Scatter Plot C ☐ Scatter Plot D
Scatter Plot(s) with a Non-Linear Association	☐ Scatter Plot A ☐ Scatter Plot B ☐ Scatter Plot C ☐ Scatter Plot D

Similar to the slope of a line, bivariate numerical data is said to have a positive association if as the x-values increase, the y-values also increase. If as the x-values increase, the y-values decrease, then the data is said to have a negative association.

b. Work with your partner to complete the table for the scatter plots in the Warm-Up.

Scatter Plot(s) with a Positive Association	☐ Scatter Plot A ☐ Scatter Plot B ☐ Scatter Plot C ☐ Scatter Plot D
Scatter Plot(s) with a Negative Association	☐ Scatter Plot A ☐ Scatter Plot B ☐ Scatter Plot C ☐ Scatter Plot D
Scatter Plot(s) with an Association That Is Neither Positive nor Negative	☐ Scatter Plot A ☐ Scatter Plot B ☐ Scatter Plot C ☐ Scatter Plot D

If the points on the scatter plot closely follow a line or a curve, the association is described as strong. If the points on the scatter plot are more spread out, the association is described as weak.

c. Work with your partner to complete the table for the scatter plots in the Warm-Up.

Scatter Plot(s) with a Strong Association	☐ Scatter Plot A ☐ Scatter Plot B ☐ Scatter Plot C ☐ Scatter Plot D
Scatter Plot(s) with a Weak Association	☐ Scatter Plot A ☐ Scatter Plot B ☐ Scatter Plot C ☐ Scatter Plot D

11.3.4 Guided Practice: Constructing Scatter Plots

1. The table shows the number of hits and home runs for professional baseball players on the Tampa Bay Rays who had at least 45 hits in the 2019 Major League Baseball (MLB) regular season.

Hits	Home Runs
154	33
80	17
107	19
155	21
82	14
138	20
86	16
135	20
102	14
55	3
46	5

a. What range of data values need to be represented along each axis to represent this data set?

Range of the Number of Hits	
Range of the Number of Home Runs	

b. Discuss with your partner which variable should be represented along each axis when constructing a scatter plot to represent this data.

c. **Construct a scatter plot to represent the data by completing the following.**

- **Label the axes with the appropriate variables.**
- **Identify and label an appropriate scale for each axis on the grid.**
- **Plot a point for each pair of values in the data set.**
- **Title the graph.**

d. Compare scatter plots with your partner. Discuss and resolve any differences, if necessary.

e. Work with your partner to describe the relationship between hits and home runs based on the data from the Tampa Bay Rays. Use at least two words from the word bank in your description.

decrease(s)	increase(s)	linear
non-linear	positive	negative

f. Work with your partner to explain why a scatter plot better represents this data set than a ine graph.

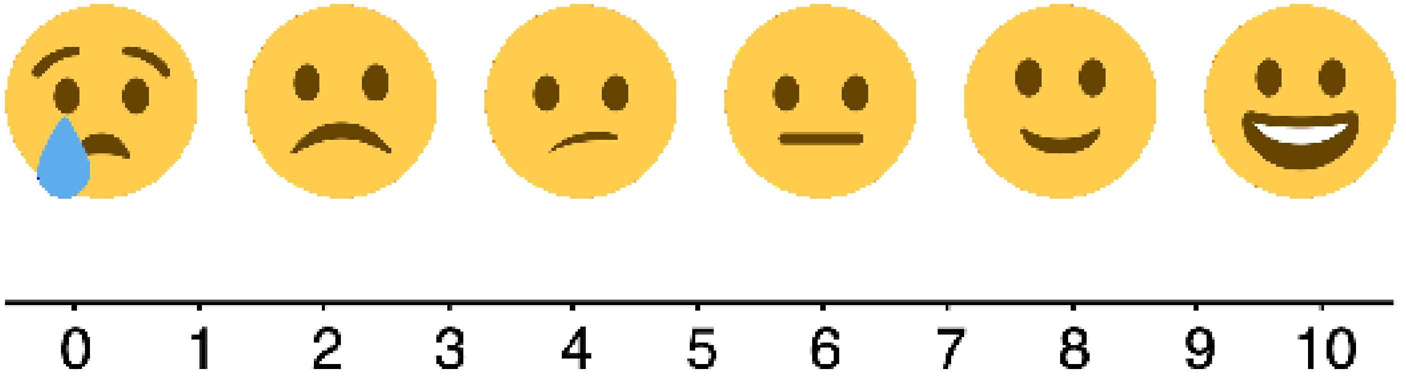
g. Ask two classmates for the explanation they wrote. Record their explanations and write your summary of their reasons. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Explanation	My Summary of Their Reason	Initials

h. Discuss the explanations from your classmates with your partner and if necessary, revise your explanation in Part F.

11.3.5 Wrap-Up: Reflection

1. Use the emoji scale to describe how confident you feel in constructing a scatter plot given a set of real-world data.



2. Explain your selection.

